

Winning Tickets Are Not Posterior Modes: Evidence from Posterior Support and Magnitude Controls

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Abstract

The lottery ticket hypothesis suggests that dense neural networks contain sparse subnetworks that can be trained in isolation when rewound to their original initialization. A tempting Bayesian interpretation is that these tickets correspond to posterior modes, or to sparse supports induced by posterior basins. We make this interpretation falsifiable: posterior-induced masks must match IMP tickets better than local magnitude, dense-trajectory, and pruning-process controls, not merely better than uniform random masks. Across five-seed MNIST and Fashion-MNIST sparsity sweeps, and a CIFAR-10 ResNet-20 epoch-1 rewind setting where IMP exceeds dense accuracy, SGLD, SGHMC, cyclical SGLD, SWAG, diagonal/KFAC Laplace, low-rank Hessian-plus-diagonal Laplace, exact last-layer Laplace, selected/tensor-block full-covariance Laplace, joint-group full-covariance Laplace, and subspace HMC all fail this stronger criterion. Posterior supports often beat random masks, but they remain tied to chain-start magnitude, dominated by dense magnitude, or closer to rewind controls than to IMP. Direct mode/ticket audits agree: posterior samples can match IMP tickets in logit or activation CKA while failing layer-sparsity and Hamming-distribution thresholds, and their basin count does not match ticket diversity. Mechanistic controls point instead to trajectory and pruning-process structure: IMP-only residual support recovers much of the fixed-mask training gap, while cross-seed transfer, posterior uncertainty, score-bin controls, overlap-matched process controls, and tensor-matched round-exclusion interventions do not rescue a posterior-mode account. In these settings, winning tickets are trajectory and pruning-process objects, not Bayesian posterior-mode supports. Exact dense full-network full-covariance CIFAR posterior baselines remain an open limitation.

1 Introduction

The lottery ticket hypothesis (LTH) states that a dense randomly initialized network contains sparse subnetworks that can be trained to match the dense network when reset to their original initialization [4]. This result raises a basic question about the object discovered by iterative magnitude pruning (IMP): is a winning ticket a combinatorial artifact of the optimization trajectory, or does it reflect a deeper probabilistic structure of the posterior over weights?

A Bayesian reading is attractive. SGD and its noisy variants have long been connected to approximate Bayesian inference [16, 11]. Bayesian deep learning further emphasizes that useful neural network solutions are distributed across broad posterior regions rather than isolated points [17, 7]. If winning tickets correspond to posterior modes or posterior-basin supports, LTH would be evidence for a selection view of learning: pruning would reveal posterior structure already present in the trained model.

This paper tests that interpretation directly. We ask:

Do posterior samples or posterior basins induce sparse supports that explain IMP winning-ticket masks beyond dense magnitude, initialization magnitude, rewind magnitude, chain-start magnitude, and pruning-process controls?

The paper contributes three pieces of evidence. First, it states the posterior-mode explanation as a support-equivalence claim with operational gates: posterior-induced supports must beat local magnitude controls, not just random masks, and mode/ticket distributions must match in both mask and function space. Second, it evaluates these gates across repeated MNIST/Fashion-MNIST sweeps and CIFAR-10 ResNet-20

posterior approximations, including stochastic samplers, Laplace variants, SWAG, subspace HMC, and direct mode/ticket distribution probes. Third, after the posterior account fails these gates, it tests a mechanistic alternative: IMP tickets are better explained by dense-trajectory and pruning-process structure, especially the IMP-only residual support.

The conclusion is negative but scoped. Posterior masks beat random masks, but they track the magnitude support of the dense checkpoint or the chain start. When chains move, they move away from the chain start without moving toward the IMP ticket. When function-space CKA is high, mask-distribution and basin-count criteria still fail. When process controls are added, the strongest positive signal is the IMP-selected residual, not a posterior-mode support.

2 Related Work

Lottery tickets and pruning. LTH was introduced by Frankle and Carbin [4]. Follow-up work connected LTH to rewinding and linear mode connectivity [5]. Random weighted networks with learned masks further emphasized the selection view of learning [13]. Single-shot pruning methods such as SNIP [9] and SynFlow [15] provide important non-Bayesian controls.

Mode connectivity and subspaces. Low-loss connectivity between neural network solutions challenges the view that trained models are isolated modes [6, 2]. For lottery tickets specifically, Paul et al. [12] argue that IMP masks encode axial subspaces that intersect matching linearly connected regions, which is a direct non-Bayesian competitor to a posterior-mode explanation. Permutation symmetries and re-basin methods further complicate parameter-space mode counting [3, 1]. Our experiments therefore include function-space clustering and linear barrier probes in addition to mask overlap.

Bayesian neural networks. SGLD provides a simple route from SGD to approximate posterior sampling [16]. SGD itself can be interpreted as approximate Bayesian inference in a stationary regime [11]. Practical posterior approximations such as SWAG [10] and exact small-model sampling studies [7] show that posterior geometry is structured and nontrivial. PAC-Bayesian and Bayesian LTH analyses [14, 8] motivate the need for controls: Bayesian language alone does not imply that IMP masks are posterior-mode supports.

Our contribution is therefore not a new posterior approximation or a claim that Bayesian inference is uninformative for pruning. It is a controlled support-equivalence test: if posterior modes explain winning tickets, their sparse supports should match IMP tickets after conditioning on dense magnitude, rewind magnitude, trajectory magnitude, and pruning-process alternatives.

3 Operational Test

Let m_{IMP} denote the mask found by IMP at a fixed sparsity. For a posterior sample $\theta^{(s)}$, the simplest posterior-induced mask is

$$m^{(s)} = \text{TopK}(|\theta^{(s)}|),$$

with the same number of kept weights as m_{IMP} . We also construct aggregate posterior masks from posterior mean, RMS, signal-to-noise ratio, high variance, and low variance.

We compare all masks by support Jaccard similarity to m_{IMP} :

$$J(m, m_{\text{IMP}}) = \frac{|m \cap m_{\text{IMP}}|}{|m \cup m_{\text{IMP}}|}.$$

The key controls are random same-sparsity masks, initialization magnitude masks, dense trained magnitude masks, rewind magnitude masks, chain-start magnitude masks for posterior chains, SNIP/SynFlow masks, fixed-mask retraining controls, channel-aligned direct mode/ticket controls, and process-residual interventions. The positive posterior-mode interpretation must satisfy a stronger condition than beating random: posterior masks must explain IMP beyond the chain-start, dense, rewind, trajectory, and process controls, and posterior samples must move support in a way that increases IMP alignment.

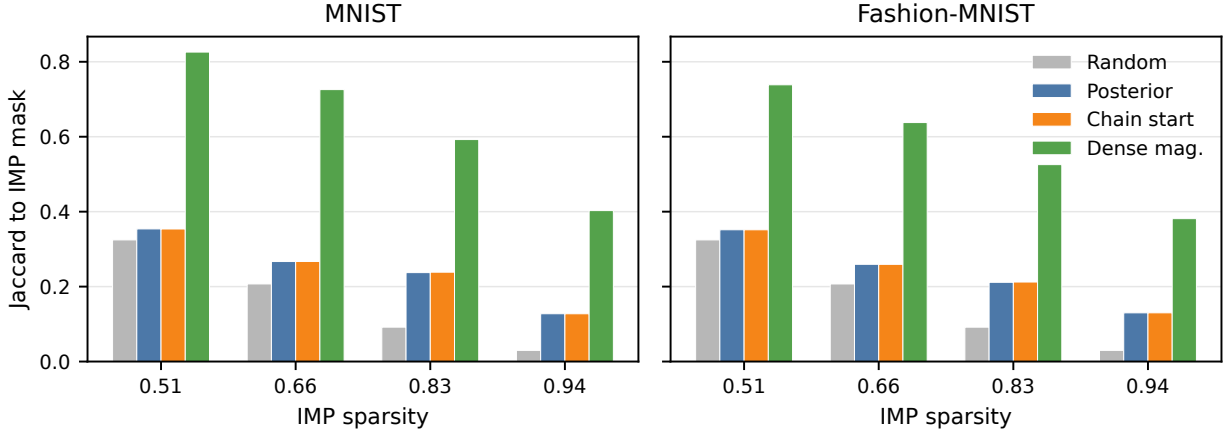


Figure 1: Gate1 control comparison on MNIST and Fashion-MNIST. Posterior masks beat random masks, but they track chain-start magnitude and remain far below dense magnitude across sparsities.

4 Current Results

4.1 Random Is Too Weak a Baseline

Tables 1 and 2 show the current Gate1 sparsity sweeps. Each row aggregates five seeds, three independent dense-start SGLD chains per seed, and posterior masks at the matched IMP sparsity.

Table 1: MNIST 5-seed Gate1 sparsity sweep. Posterior masks beat random but fail against stronger controls at every sparsity. “Post-Chain” is the Jaccard overlap between posterior masks and their chain-start magnitude masks.

Config	Runs	Sparsity	Posterior	Random	Chain Start	Post-Chain	Dense Mag	Gate1
r2 p0.30	5	0.5100	0.3539	0.3246	0.3537	0.9303	0.8259	Fail
r3 p0.30	5	0.6570	0.2669	0.2071	0.2670	0.9115	0.7258	Fail
r5 p0.30	5	0.8319	0.2373	0.0918	0.2385	0.9298	0.5924	Fail
r8 p0.30	5	0.9424	0.1275	0.0298	0.1275	0.9423	0.4028	Fail

Table 2: Fashion-MNIST 5-seed Gate1 sparsity sweep. The failure mode matches MNIST. At the highest sparsity, IMP accuracy drops below the dense model, so that row should be interpreted as a high-sparsity stress case.

Config	Runs	Sparsity	Posterior	Random	Chain Start	Post-Chain	Dense Mag	Gate1
r2 p0.30	5	0.5100	0.3516	0.3246	0.3516	0.9314	0.7385	Fail
r3 p0.30	5	0.6570	0.2595	0.2070	0.2594	0.9160	0.6378	Fail
r5 p0.30	5	0.8319	0.2114	0.0917	0.2122	0.9212	0.5255	Fail
r8 p0.30	5	0.9424	0.1300	0.0297	0.1300	0.9408	0.3813	Fail

Posterior masks beat random support at every tested MNIST/Fashion-MNIST sparsity, but the random gap is explained by chain-start magnitude. On MNIST and Fashion-MNIST, posterior-minus-chain-start gaps stay near zero while posterior-to-chain-start overlap remains around 0.91–0.94. Dense magnitude is much closer to IMP, with dense-minus-posterior gaps of 0.275–0.472 on MNIST and 0.251–0.387 on Fashion-MNIST. The same pattern holds in the generated mode-distribution audit: posterior supports beat uniform random masks in 58 of 59 grouped comparisons, but none of the 59 posterior-versus-chain-start comparisons

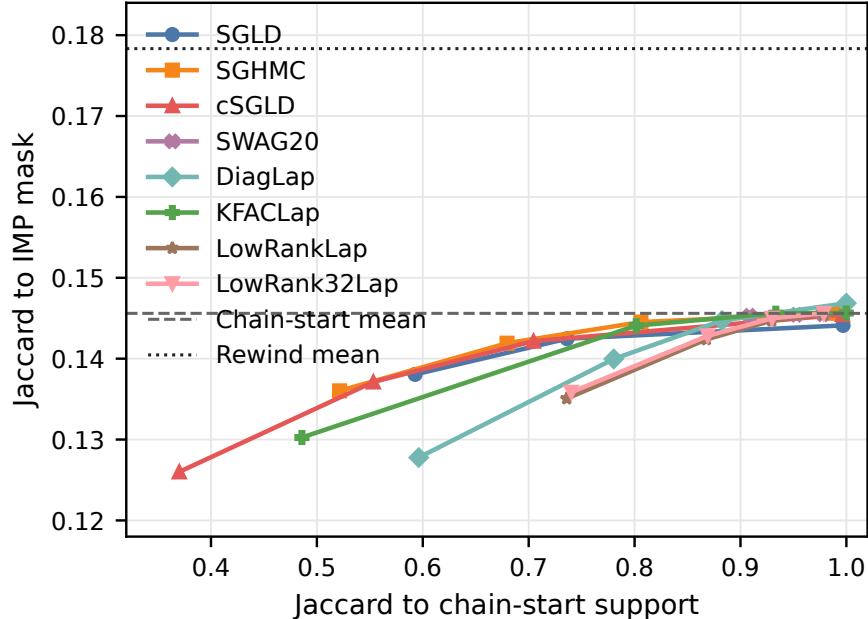


Figure 2: CIFAR-10 ResNet-20 epoch-1 rewind movement diagnostics. Across SGLD, SGHMC, cyclical SGLD, full-network SWAG, diagonal Laplace, KFAC-style Laplace, and low-rank Hessian-plus-diagonal Laplace, moving support away from the chain start does not increase overlap with the IMP mask.

beats the matched chain-start support by more than 0.005 Jaccard: 43 are practically tied to the control, 15 favor chain-start magnitude, and one is mixed. Rewind magnitude is closer to the IMP ticket than posterior support in 55 of 57 grouped comparisons.

4.2 Sampler Movement Does Not Point Toward Tickets

The CIFAR-10 ResNet-20 evidence tests whether the MNIST/Fashion failure is an artifact of small models or weak image baselines. In the long-budget epoch-1 rewind row, dense accuracy is about 88.5% and IMP reaches about 89.8% at 83.2% sparsity, yet SGLD, SGHMC, cyclical SGLD, SWAG, diagonal Laplace, KFAC-style Laplace, low-rank Hessian-plus-diagonal Laplace, and subspace HMC all fail the same control. Increasing sampler scale often lowers posterior-to-chain-start overlap, but posterior-to-IMP overlap does not improve and the epoch-1 rewind support remains closer to IMP.

The negative movement result is broad across posterior families. The sampler grid includes dense-start and independent-start multi-chain SGLD, SGHMC, cyclical SGLD, a 20-snapshot full-network SWAG movement row, diagonal and KFAC-style Laplace, rank-16, rank-32, rank-64, and rank-128 top-Hessian low-rank Laplace movement rows, exact final-head Laplace, selected and tensor-block-diagonal full-covariance Laplace, joint-group Laplace, and low-dimensional HMC in random, trajectory, and top-Hessian subspaces. Full-batch HMC and an exact dense full-network Laplace sanity check on small digits MLPs validate higher-fidelity small-model code paths while preserving the same qualitative support failure.

4.3 Richer Covariance Still Fails the Support Gate

The covariance objection is the strongest bounded limitation. We therefore push exact covariance as far as the workstation budget permits. We use a 22,064-parameter block-diagonal probe covering 22,064 trainable weights; at scale 10^{-4} sample accuracy remains 88.1%, global posterior-to-chain-start support falls to 0.8287, the selected-block posterior-to-IMP support is below the selected-block chain-start control by 0.0114 on average, the global posterior-minus-chain gain is only 0.0036, and rewind remains closer by 0.0292. Thus, this 22,064-parameter block-diagonal row moves support but not toward tickets.

Sampler	LR/Scale/Step	Dense	IMP	Post.	Chain	Post-chain	Rewind
SGLD	10^{-6}	0.8845	0.8993	0.1425	0.1441	0.7362	0.1784
SGLD	3×10^{-6}	0.8845	0.8993	0.1381	0.1441	0.5928	0.1784
cSGLD	10^{-6}	0.8852	0.8960	0.1422	0.1454	0.7046	0.1789
DiagLap	10^{-3}	0.8849	0.8980	0.1447	0.1469	0.8826	0.1787
DiagLap	10^{-2}	0.8849	0.8980	0.1278	0.1469	0.5961	0.1787
KFACLap	10^{-2}	0.8860	0.8956	0.1303	0.1456	0.4859	0.1775
LowRank32Lap	10^{-2}	0.8853	0.8964	0.1358	0.1457	0.7402	0.1795
LowRank64Lap	10^{-2}	0.8874	0.8954	0.1339	0.1433	0.7397	0.1766
LowRank128Lap	10^{-2}	0.8867	0.8967	0.1351	0.1453	0.7358	0.1780
SWAG20	64	0.8859	0.8973	0.1453	0.1455	0.9086	0.1782
RandSubHMC	3×10^{-3}	0.8866	0.8965	0.1440	0.1440	0.9766	0.1779
TrajSubHMC	10^{-3}	0.8849	0.8961	0.2290	0.2292	0.9915	0.1792
Hess16SubHMC	3×10^{-4}	0.8873	0.8974	0.1468	0.1468	0.9994	0.1799
Hess32SubHMC	3×10^{-4}	0.8868	0.8973	0.1461	0.1461	0.9993	0.1788

Table 3: Representative five-seed CIFAR-10 ResNet-20 epoch-1 rewind movement diagnostics at r5 p0.30. Posterior support either remains near the chain-start magnitude support or moves away from it, but posterior-to-IMP overlap (Post.) does not exceed the matching chain-start control. TrajSubHMC is higher because its chain-start control is the matched dense trajectory magnitude support; the full generated movement summary is in the supplemental generated movement table.

Wider exact covariance rows give the same answer, and this 68,144-parameter block-diagonal row, covering 68,144 trainable weights, reports that sample accuracy remains 88.0%, global support falls further to 0.7400, selected-block posterior-minus-chain remains negative at -0.0050, global posterior-minus-chain gain shrinks to 0.0010, and rewind remains closer by 0.0319, and this 68,144-parameter block-diagonal row remains negative. The 68,144-parameter joint-group row keeps the same 68,144 trainable weights while including cross-tensor covariance inside groups; sample accuracy remains 88.1%, global posterior-to-chain-start support falls to 0.7148, block posterior-minus-chain remains -0.0050, global posterior-minus-chain is 0.0015, and rewind remains closer by 0.0311, so this 68,144-parameter joint-group row also fails. The next row is this 86,576-parameter joint-group row, covering 86,576 trainable weights, and it preserves sample accuracy at 88.3%; global posterior-to-chain-start support is 0.7863, block posterior-minus-chain stays negative at -0.0023, global posterior-minus-chain stays 0.0006, and rewind remains closer by 0.0317. The largest row is this 270,896-parameter joint-group row and covers all 270,896 weight parameters; sample accuracy remains 88.2%, global posterior-to-chain-start support is 0.7389, block posterior-minus-chain remains negative at -0.0019, global posterior-minus-chain is -0.0019, and rewind remains closer by 0.0362. These rows remove the practical weight-tensor coverage objection, while literal dense CIFAR full-network covariance remains outside budget: a single float64 dense matrix would require 553.1 GiB, and matrix plus Cholesky would require 1,106.3 GiB.

4.4 Function Space Agreement Is Not Mask Equivalence

We also make the original proposal’s distributional criterion explicit. On digits, posterior sample masks fail two of the four stated thresholds: layer-sparsity KS has $p = 0.0788$ and pairwise mask-Hamming overlap is 0.631, while logit CKA passes. Mean-shift clustering in raw parameter PCA collapses posterior samples to one representative mode; the raw-parameter basin entropy is 0.0 nats with effective cluster count 1.0. The full-data CIFAR-10 ResNet-20 strengthens this separation: posterior samples can keep logit/final-hidden activation CKA above 0.93/0.91 while failing layer-sparsity, Hamming, or posterior-basin-count criteria.

Alignment and stronger direct posterior probes do not change this conclusion: activation-channel alignment are insufficient. The activation-aligned CIFAR row still fails layer KS and Hamming while retaining high activation CKA. Weight-correlation channel alignment reaches the same conclusion. The saved artifact rerun supports record-level post-hoc matching, and a block-coordinate ResNet channel audit still leaves global-channel posterior/ticket Hamming near 0.21 rather than ticket-like agreement. An exact stage-1 enumeration validates the local solver on a small ResNet subgraph, while exhaustive full-data graph isomorphism

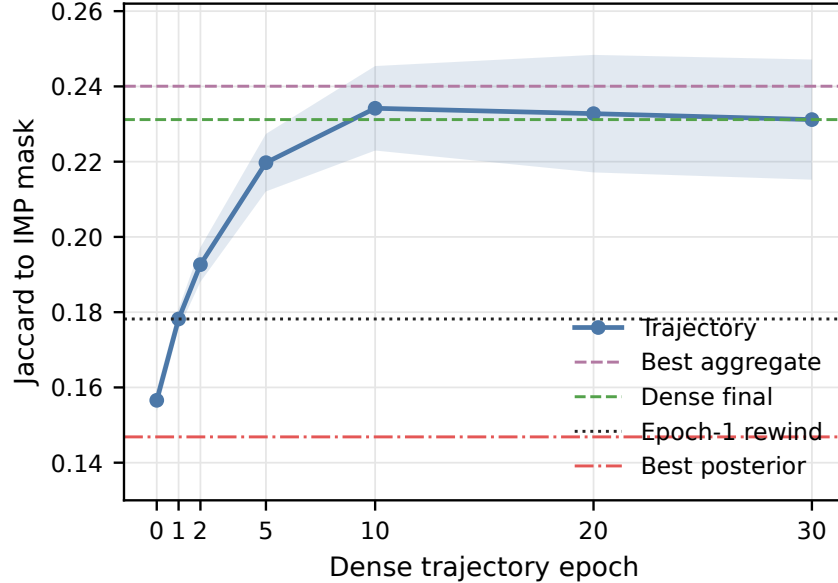


Figure 3: CIFAR-10 ResNet-20 dense-trajectory controls. Magnitude supports along the matched dense trajectory, especially around epochs 10–30, and the best aggregate trajectory magnitude score are closer to the IMP mask than any global posterior support in the movement diagnostics.

remains infeasible.

The multi-chain and high-fidelity direct rows address the concern that a single local chain was too weak. A dense-start direct probe with 75 cyclical-SGLD posterior samples fails layer KS and Hamming while maintaining high CKA; an independent-start version with 15 dense chain starts and 75 samples does the same. The rank-128 Hessian-plus-diagonal direct probe is the partial exception: 50 posterior samples now pass the Hamming-overlap threshold (0.816) and retain high logit/final-hidden activation CKA (0.932/0.910), but still fail the layer-sparsity KS threshold ($p = 2.0 \times 10^{-6}$), and all 50 samples still collapse to one parameter-PCA basin. A streamed exact joint-group direct probe over the same all 270,896 weight parameters closes the full-weight Gaussian gap: 25 joint-group posterior samples keep sample accuracy remains 88.3%, posterior-to-chain-start Hamming is 0.0503, layer-sparsity KS is $p = 1.1 \times 10^{-8}$, Hamming overlap is 0.000, logit/final-hidden activation CKA remains 0.937/0.920, and All 25 joint-group samples still collapse to one parameter-PCA basin.

4.5 IMP Residuals Are Process Objects

Posterior failure leaves a positive mechanism to explain. Dense trajectory magnitude is useful: in the matched CIFAR dense-trajectory probe, trajectory magnitude supports are closer to IMP than posterior supports, and fixed-mask training shows that final dense, epoch-10, and RMS trajectory masks are much better than random. They are still below IMP, so the remaining question is what IMP adds.

Residual swap and removal-order controls localize the gap to a process-selected IMP-only residual. Anatomy, predictor-mask, cross-seed, direct-transfer, base-compatibility, posterior-ordering, and stratified controls give the same localization. Learned mask baselines do not replace that residual: Gem-Miner-style, Bernoulli/Concrete variational pruning, and the hard-concrete L0 gate baseline are below IMP in the current CIFAR support/accuracy rows and do not rescue the calibration/OOD tradeoff. Their learned objective can change uncertainty behavior, but not into a ticket-support explanation.

The strongest process controls test whether simpler score composition explains the residual. A tensor matched excluded oracle is still below the round-selected IMP residual, and a tensor+score matched excluded oracle remains below it even after matching parameter tensor and within-tensor round-score deciles. A residualized round-score projection shows that removing base, dense-final, and final-IMP magnitude subspaces re-

duces both accuracy and oracle overlap. The original posterior-projection round-selected row reaches 88.47% accuracy. A posterior-residualized row reaches 88.25% after additionally residualizing against diagonal-Laplace posterior RMS, posterior standard deviation, and posterior RMS-minus-dense features; the gap is 0.23 percentage points, 5/5 positive seeds with 95% CI $[-0.03, 0.48]$, and overlap drops from 0.6773 to 0.4850, a 0.1923 paired drop with 5/5 positive seeds. The diagonal-Laplace posterior score subspace does not explain away the process signal. Finally, the learned trajectory/process subspace check gives the cleanest compact result: the round-selected row reaches 88.69% accuracy, the learned-subspace residualized row reaches 88.21%, the gap is 0.48 percentage points, 5/5 positive seeds with 95% CI $[0.29, 0.67]$, and oracle overlap falls from 0.6807 to 0.4917, a 0.1890 learned-subspace paired drop. Thus the current learned process subspace captures broad directions but does not substitute for the particular residual coordinates selected by IMP.

5 Discussion

These results constrain a simple Bayesian interpretation of lottery tickets. The posterior signal exists in a weak sense: posterior-induced masks are not random. But this is not enough. The same signal is already present in the dense, rewind, trajectory, or chain-start magnitude support. The explicit distribution-equivalence audit turns this into a negative result rather than a visual or single-metric impression: posterior support distributions separate from random masks, yet they are tied to or worse than local magnitude controls across the posterior families tested here. The evidence therefore favors a trajectory/magnitude subspace account plus a process-selected residual over a posterior-mode account.

This conclusion is compatible with mode-connectivity results. Independent solutions can occupy separated regions in parameter or function space, and yet the sparse supports induced by local posterior samples need not explain the IMP ticket. A generated linear connectivity audit makes this point concrete. On MNIST and Fashion-MNIST, dense-to-IMP linear barriers are nearly zero (0.0026 and 0.0395), but posterior support remains slightly below the chain-start magnitude control. On CIFAR-10 ResNet-20, the long SGLD and SWAG dense-to-IMP barriers are large (3.0827 and 3.7402), but posterior support is still tied to chain-start support. Thus linear barriers are orthogonal landscape diagnostics, not evidence of posterior-ticket equivalence.

Scope of the claim. The claim is not that Bayesian neural-network posteriors are useless, nor that no posterior approximation could ever correlate with IMP coordinates. The claim is narrower: under the posterior families and CIFAR-10/MNIST/Fashion-MNIST settings tested here, posterior-induced sparse supports do not satisfy the support-equivalence claim once dense, rewind, trajectory, and pruning-process controls are included. The positive evidence in the paper is the alternative localization: useful ticket structure is better explained by trajectory magnitude plus a process-selected IMP residual.

6 Limitations and Next Experiments

The evidence is broad, but the paper should still be read as a controlled negative result under the posterior approximations we can currently test, not as a theorem about all Bayesian neural-network posteriors. The main limitations are:

- alignment robustness. Activation-channel alignment, weight-correlation channel alignment, record-level post-hoc matching, a block-coordinate global channel audit, and exact stage-1 enumeration are negative or bounded, but exhaustive full-data graph isomorphism remains infeasible and unimplemented;
- full-network exact/full-covariance Laplace or another still broader high-fidelity posterior baseline. Current evidence includes multi-chain cyclical SGLD, rank-128 low-rank Laplace, all-weight joint-group Laplace, 20-snapshot SWAG, low-rank Hessian-plus-diagonal Laplace, tensor-block and joint-group full-covariance probes, and subspace HMC, but the exact dense full-network CIFAR posterior remains absent;

- broader learned-mask distribution variants. Gem-Miner-style score training, Bernoulli/Concrete variational pruning, and hard-concrete L0 gates do not rescue support, accuracy, calibration, or CIFAR-100 OOD behavior in the current five-seed rows, but additional permutation-aware learned-mask distributions would make this negative claim cleaner;
- additional process-causality controls. The current residual-swap, anatomy, predictor-mask, transfer, base compatibility, ordering, stratification, IMP-process, tensor+score-matched round-exclusion, residualized-score projection, posterior-residualized, and learned-subspace residualized controls do not exhaust the possible causal interventions on the pruning process;
- reproducibility hardening. The repository has a dependency list, checked environment lock, Makefile targets, artifact verifier, CI workflow, public release manifest, local release archive, source-only repository snapshot, CPU artifact-verification container, GPU training-container definition, external-validation runbook, and submission handoff. It still lacks public archive, public repository, external CI, and external GPU-run receipts.

Since the aligned, multi-chain, low-rank Laplace, and block or joint-group Laplace CIFAR runs preserve the current pattern, the next decisive empirical gap is an even stronger high-fidelity posterior baseline. If that baseline also stays near the chain-start/rewind controls, the negative result is the right paper: posterior samples reveal local magnitude structure, not IMP ticket structure. If it contradicts the current pattern, the claim should be weakened to a partial-equivalence statement that isolates the architectures, sparsities, and posterior approximations where the Bayesian account survives.

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